Generative AI Challenge: Verifiable AI Agents with Hologram

Ser ANDI AI Scholarship · NODO EAFIT · Verana Foundation

verana.io | hologram.zone

The Problem

Who do you trust when you chat with a bot?

- Current chatbots live on centralized platforms
- You can’t know who operates them or what they do with your data
- There is no way to verify whether an AI agent is legitimate
- Anyone can create a bot that pretends to be a company or a person

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The Solution: Verana + Hologram

A trust layer for AI agents

Verana

Open infrastructure for decentralized identity (DIDs + W3C Verifiable Credentials)

Hologram

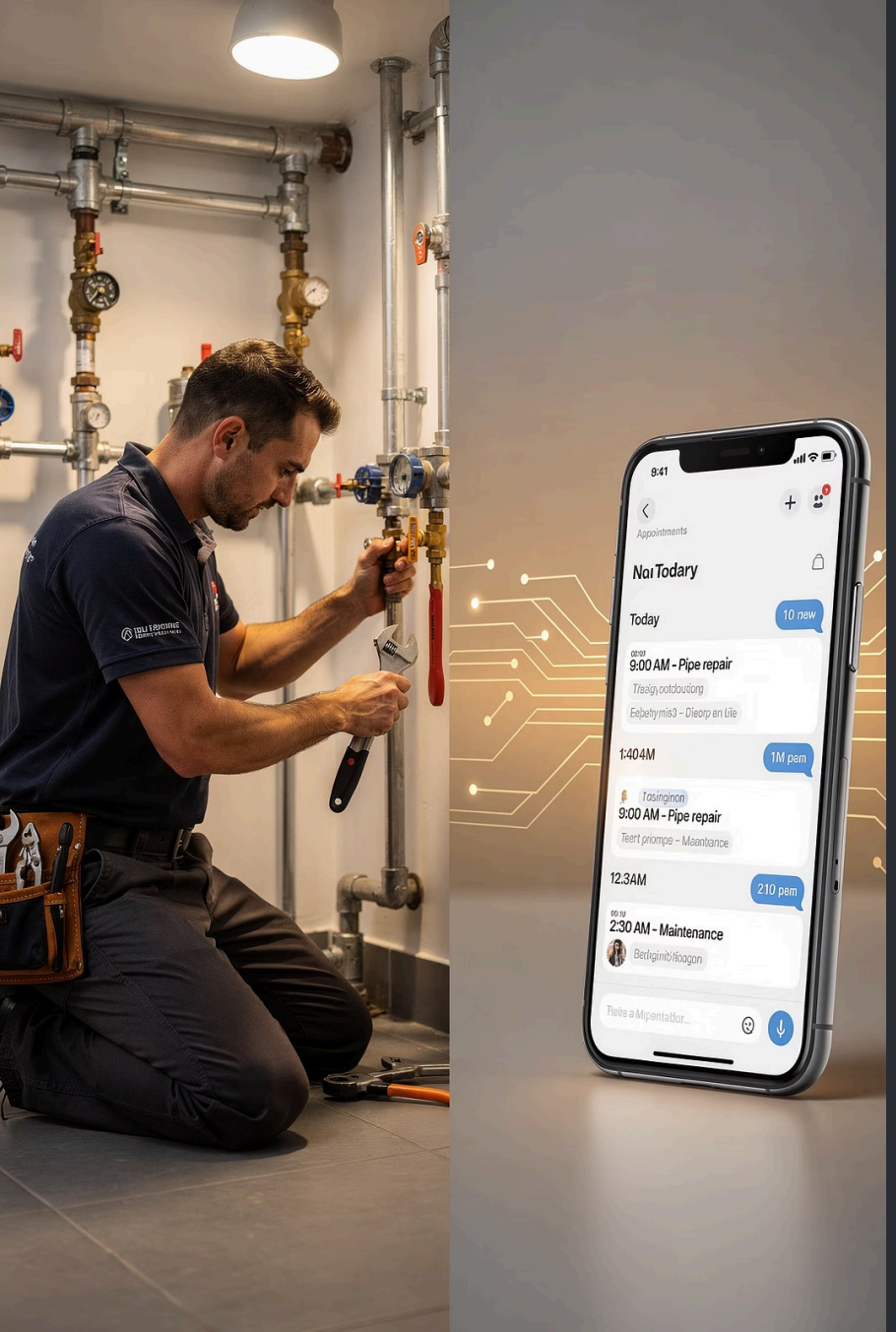
Messaging app where each bot has a verifiable identity

Confirmation

Before chatting, the user can confirm **who operates the agent**

Verification

Everything is cryptographically backed — you don't have to "trust" it, you can **verify** it



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Your Challenge

Build a platform to create Persona AI Agents

An **Persona AI Agent** represents a person and acts on their behalf.

Example: a plumber has an AI agent that manages his calendar. Customers chat with the agent in Hologram to schedule an appointment.

Your mission: create a web app where anyone can create, configure, and publish their own Persona AI Agent — without writing code.

How You'll Get There: 3 Steps

Progressive learning: from simple to complex

01

Step 1 — Learn (20%)

Set up and run an AI chatbot with Hologram on your machine

02

Step 2 — Learn (15%)

Deploy your bot on Kubernetes (publicly accessible)

03

Step 3 — The Challenge (40%)

Build the web platform to create Persona AI Agents

The `challenge.md` document has the detailed step-by-step instructions.

Step 1: Your First Chatbot

Fork → Set Up → Run → Chat in Hologram

1. Fork the repo `verana-labs/verana-eafit`
2. Configure environment variables (LLM provider, prompt, languages)
3. Customize the `agent-pack.yaml` (name, personality, RAG)
4. `docker compose up` — runs everything locally
5. Expose with **ngrok** and connect from **Hologram**

Technologies: Docker, Node.js, LLMs (OpenAI/Ollama), RAG, VS Agent

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Step 2: Cloud Deployment

From your laptop to the real world with Kubernetes

1. Receive `kubeconfig` credentials for your namespace
2. Customize the YAML manifests in `k8s/`
3. Run the deployment script
4. Your bot ends up at: `miagente.tunombre.eafit.testnet.verana.network`

You no longer need ngrok — your bot is online 24/7.

Step 3: The Web Platform

The main challenge: an app to create bots without code

The web app must allow:

Login

For users (optional: Google OAuth)

Create

A new Persona AI Agent with a visual form

Configure

Prompt, MCP tools, RAG, persona data, and service

Publish/Unpublish

The bot in Kubernetes with one click

Direct Link

To the bot so it can be scanned with Hologram

Include at least **2 MCP servers** (e.g., Google Calendar, X/Twitter)

Tools and Support

Everything you need to get started

Tool	What for
VS Code / Windsurf / Cursor	Writing code
Git + GitHub	Version control and collaboration
Node.js + pnpm	Runtime and dependencies
Docker	Running services locally
ngrok	Public tunnel for development
Hologram	Testing your chatbot
kubectl	Deploying to Kubernetes

Support: Verana Discord → channel **#eafit-challenges** https://discord.com/invite/edjaFn252q

Verana's technical mentor will support you throughout the entire process.

Evaluation and Next Step

How you will be evaluated

Criterion	Weight
Step 1 — Chatbot working	20%
Step 2 — k8s deployment	15%
Step 3 — Web application	40%
MCP integrations	10%
Code + Docs + Creativity	15%

Bonus (+5% each)

- More MCP servers
- Real RAG
- Exceptional UI
- Tests
- Video demo

Next step: Read the `challenge.md` document, install the tools, and join the Discord. See you in #eafit-challenges!